

SIMATS ENGINEERING

ASSESSMENT TOOL - II : DESIGN TASK

Course Code	ECA14
Course Name	Embedded Systems
Course Outcome	CO1
Assessment Tool	Design Task
Question No.	19
Question Title	19. Design a complete CAN-based Smart Home Automation System for a 3-floor residence with 60 control points including lighting, HVAC, Security and entertainment subsystems. Apply formal design methodology to specify the networked embedded architecture, message priority assignment, and bus-load distribution. Demonstrate how system achieves sub-200ms response latency for normal controls.
Maximum Marks	100
Bloom's Taxonomy Level	BTL 5

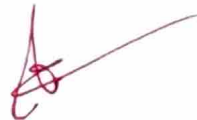
Evaluation Rubric (100 Marks)

Criteria	Weightage	Level 4	Level 3	Level 2	Level 1	Marks Awarded
Understanding of Design	20	☒ 20	☐ 16	☐ 12	☐ 8	20
Evaluation of Design	25	☒ 25	☒ 20	☐ 15	☐ 10	20
Justification	20	☒ 20	☒ 16	☐ 12	☐ 8	16
Suggestions for Improvement	20	☐ 20	☒ 16	☐ 12	☐ 8	16
Written Communication	15	☐ 15	☒ 12	☐ 9	☐ 6	12
Total	100					84/100

Code of Conduct:

I M. Tahavi/192411196 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Problem Statement:

Design a CAN (Controller Area Network) based Smart home Automation System for a 3-floor house with 60 Control points.

Subsystems:

- Lighting - 24 points
- HVAC (Heating, Ventilation & Air) - 12 points
- Security - 12 points
- Entertainment - 12 points

Total - 60 Control points

2. Formal Design Methodology

Step 1: Requirement analysis

- 3-floor smart home
- 60 networked devices
- Reliable Communication
- Real-time operation
- Fire alarm response $< 1\text{ms}$
- Normal Control response $< 200\text{ms}$
- Low power Consumption
- Fault tolerance
- Easy expansion

Step 2: System Specification

• Communication Protocol: CAN Bus (CAN 2.0B)

• Bus Speed: 1Mbps

• MCU: STM32 / ESP32 with CAN Controller

• Power Supply: 24V DC

Sensors:

• Motion

• Smoke

• Temperature

• Door

• Gas

Actuators:

• Lights

• Fan

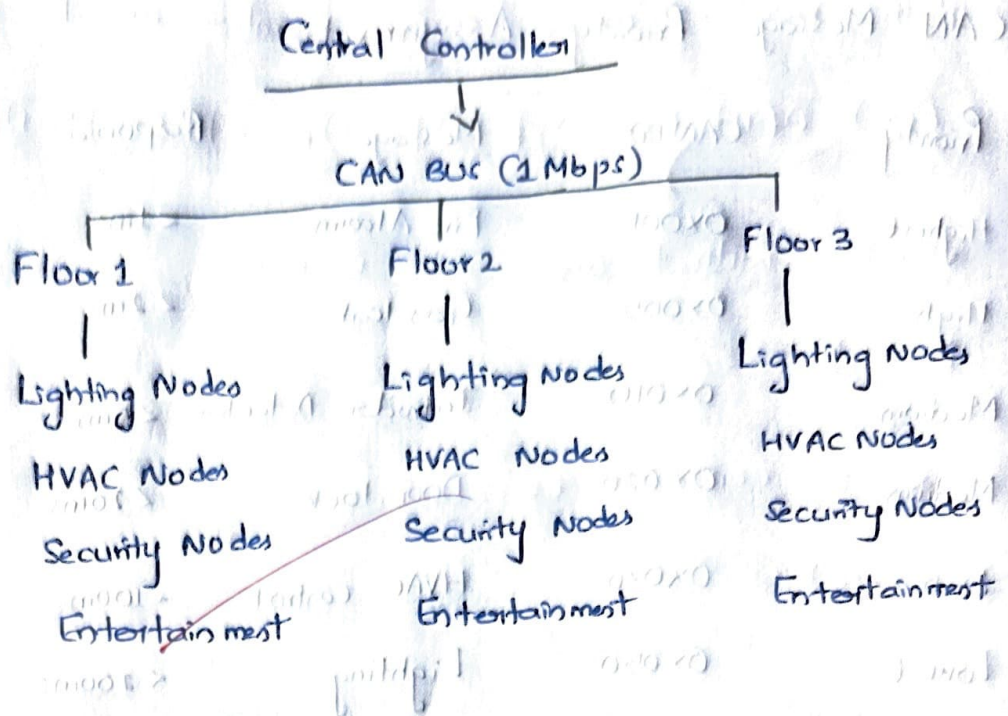
• AC

• Door lock

• Alarm Siren

• TV / Entertainment Controller

Step 3: Hardware Architecture



Embedded Network Architecture:

Each floor contains:

- 8 Lighting Nodes
- 4 HVAC Nodes
- 4 Security Nodes
- 4 Entertainment Nodes

Each node consists of:

- Microcontroller
- CAN Transceiver
- Sensor / Actuator
- Power Supply

All communicate over a CAN bus.

CAN Message Priority Assignment:

Priority	CAN ID	Message	Response
Highest	0x001	Fire Alarm	<1ms
High	0x005	Gas leak	<2ms
Medium	0x010	Intruder Detection	<5ms
Medium	0x020	Door lock	<20ms
Low	0x030	HVAC control	<100ms
Lowest	0x050	Lighting	<200ms

CAN arbitration ensures lower CAN IDs have higher priority.

BUS Load Distribution:

Sub System	Nodes	Traffic
Lighting	24	30%
HVAC	12	20%
Security	12	35%
Entertainment	12	15%

Total Bus Utilization = 60-70%

This is below the recommended 80% preventing congestion.

Timing Analysis

Normal Control

Lighting Switch

Transmission = 40ms

CAN arbitration = 10ms

Controller Processing = 50ms

Actuator Delay = 70ms

Total = 170ms ($< 200ms$)

Requirement Satisfied

Fire Alarm

Smoke Sensor Detects Fire



Highest priority CAN ID



CAN arbitration



Immediate Broadcast



Alarm + Unlock Exit Door + Notify Controller

Total Delay:

Sensor Detection = 0.2ms

CAN Transmission = 0.2ms

Processing = 0.3ms

Alarm Activation = 0.2ms

Total = 0.9ms ($< 1\text{ms}$)

Requirement Satisfied

Simulation & Validation:

- CANoe / CAN analyzer simulation
- Hardware-in-the-loop (HIL)
- Fault injection
- Smoke alarm testing
- Network traffic analysis
- Latency measurement
- Reliability testing

Conclusion:

A CAN-based distribution smart home automation system operating at 1Mbps efficiently connects 60 control points across 3 floors. By assigning highest-priority CAN IDs to safety-critical events such as fire alarm, the system guarantees sub-1ms emergency response, while routine operations like lighting, HVAC and entertainment remain within the sub-200ms response requirement.